COURSE TITLE : PAPER TECHNOLOGY
COURSE CODE : 4122
COURSE CATEGORY : B
PERIODS PER WEEK : 4
SEMESTER : 4
PERIODS PER SEMESTER : 56
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
I	Beating and Refining Consistency controllers Storage Chest and Agitators,fibre Recovery Test	5 4 4 1
II	Wet end additives Paper machine Head box Paper machine wire part Test	4 4 5 1
III	Paper Machine dry end Coating of Paper Calendaring Finishing, Types of paper Test	4 3 3 3 1
IV	Chemical recovery Causticizing Effluent treatment Air Pollution Control Test	3 4 4 2 1
	Total	**56**

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand the Beating and Refining of Pulp
	2	Understand Storage Chest and Agitators In Stock Preparation Side
	3	Comprehend Fibre Recovery
2	1	Understand Sizing ,Filling and Dyeing
	2	Understand Paper Machine Head box
	3	Comprehend Paper Machine Wire part
3	1	Understand Press part and Dryer part of Paper machine
	2	Know Coating of Paper
	3	Understand Finishing
	4	Understand Speciality Paper
4	1	Understand Chemical Recovery Operations In Paper Mills
	2	Know Caustic zing Process
	3	Understand The Need And Relevance Of Effluent Treatment And Air Pollution Control

Specific outcome

MODULE I

1.1.0 Understand the Beating and Refining of Pulp

1.1.2 Explain the equipment's and operations of batch and continuous Hydrapulpers
1.1.3 Draw the figures of the equipment's of Kneaders and deflakers and discuss their operations
1.1.4 Describe storage chest with figure
1.1.5 Describe agitators with diagram
1.1.6 State the different types of stock and machine chests
1.1.7 Explain 2 types of storage chests with figures
1.1.8 State beating and refining process
1.1.9 Describe the factors influencing beating and refining
1.1.10 State freeness and fibrillation

1.1.11 Describe the effect of freeness and fibrillation on the properties of paper
1.1.12 Explain different types of beaters & refiners such as – Hollander beater, John's betrams beater. conical refiners, disc refiners ,wide angle refiner

Understand Storage Chest and Agitators in Stock Preparation Side

1.1.1 State consistency and stock proportioning
1.1.2 State the need of consistency controlling
1.1.3 List the different types of consistency controllers and consistency transmitters
1.1.4 State the working principles of consistency controllers and stock proportions
1.1.5 List properties of paper
1.1.6 Describe various properties according to the classification
1.1.7 List the types of absorbency stock proportioners
1.1.8 Describe various absorbency stock proportioners
1.1.9 Distinguish between the high density a low density cleaning
1.1.10 Describe the theory of cleaning
1.1.11 Explain Deaerators with figure

1.2.0 Comprehend Fibre Recovery

1.2.1 State the need of Fobre recovery
1.2.2 state save all
1.2.3 List the method of fiber recovery
1.2.4 List the different types of save alls
1.2.5 Compare the following types of save all – settling and sedimentation. Conical saveall, Flotation saveall, SP Krofta
1.2.6 State the need of filtration
1.2.7 Explain vacuum – disc filter
1.2.8 State stuff box

MODULE II
2.1.0 Understand Sizing ,Filling and Dyeing

2.1.1 State sizing of paper
2.1.2 Differentiate between internal and external sizing
2.1.3 Explain different sizing mentors such as – Acidic, Alkaline and Neutral sizing
2.1.4 Discuss surface sizing methods and sizing chemicals
2.1.5 Explain the figure – straight, Horizontal, Inclined and sped sizes (Presses)
2.1.6 State the purpose of filling and loading
2.1.7 Describe the fillers and their chemical composition
2.1.8 State dyeing of paper
2.1.9 Explain Dyeing of Paper& classifications of dyes
2.1.10 Describe the general properties of above groups of dyes
2.1.11 Describe wet strength additives
2.1.12 Explain the natural & synthetic types of additives

2.2.0 Uu**nderstand Paper Machine Headbox**

2.2.1 State the function of head box
2.2.2 Describe the open and closed head box
2.2.3 Draw the head boxes and its parts such as distribution section, middle section, slice , head boxes
2.2.4 Define slices
2.2.5 Describe straight, convergent and combination slice with figures
2.2.6 Specify the relation between head of stock behind the slice and velocity
2.2.7 Calculate the velocity
2.2.8 State the role of distribution roll and high turbulence head box
2.2.9 Explain the different types of manifolds, head box over flow, head box recirculation, jet wire ratio and head calculations

2.3.0 Comprehend Paper Machine Wire part

2.3.1 State Configuration of paper machine
2.3.2 Describe the general arrangement of paper machine systems
2.3.3 Describe the operation of paper machine systems and sheet formation
2.3.4 Explain the different types of paper machine wire part such as cylinder mould, fourdriner, twin wire formers , special formers and combination with diagrams
2.3.5 Explain the different components of wire part forming medium wire weave pattern and sheet formation

MODULEIII

3.1.0 Understand Press part and Dryer part of Paper machine

3.1.1 State the different type of presses such as solid, suction, venta nip, reverse & trinip and give their uses
3.1.2 Explain the different types of felts such as woollen, synthetic, weaving and needling and state their need
3.1.3 State principles of drying
3.1.4 Explain different methods of drying such as conduction, convection and radiation
3.1.5 State t the features affecting drying such as heat content temperature and moisture content and give the method of removing them
3.1.6 Draw the different type of dryers and its arrangements such as yankee cylinder (MG) multi cylinder (MF) combined, infrared radiation , radiant drying and state their functions
3.1.7 State the different types of steam and condensate the trial system such as cascade, pressure regulated, differential pressure regulated
3.1.8 Describe the different condensate removal methods such bucket scoop siphon rotating and stationary
3.1.9 Describe the operating problems related to the dryer section
3.1.10 Describe control of the dryer section
3.1.11 Specify the formula to calculate drying capacity

3.1.12 Calculate the amount of heat transfer by using above equation (simple problems only)

3.2.0 Know Coating of Paper

3.2.1 Define coating
3.2.2 Describe coating operations
3.2.3 State advantages of coating
3.2.4 State coaters
3.2.5 List the coating chemicals and pigments
3.2.6 Describe coating chemicals and pigments
3.2.7 Describe the different types of coaters
3.2.8 State the theory of coating
3.2.9 Describe the different types of coaters, such as Brush coaters, cast coaters, blade coaters

3.3.0 Understand Finishing

3.3.1 State the importance of moisture on paper during calendaring
3.3.2 Describe different types of calendaring
3.3.3 State finishing of paper
3.3.4 State the need of different type of reels such as pope reels ,core mind reels friction four drum reels ,two drum upright reels, three drum revolving reel
3.3.5 Explain the slitting & rewinding
3.3.6 Specify the calendaring troubles
3.3.7 Explain rotary (simplex and duplex) cutting, guillotine trimming and their merits with fig
3.3.8 State the need of sorting and counting of sheets
3.3.9 Describe the method of packing of reels and sheets
3.3.10 Describe roll wrapping machine with diagram

3.4.1 **Understand speciality Paper**

3.4.2 Define converting of paper
3.4.3 State corrugating of paper
3.4.4 Describe the uses of wet and dry waxing of paper
3.4.5 Describe the laminating of paper
3.4.6 List the properties sizes etc regarding paper bags and envelops
3.4.7 Explain the special characteristics of paper and board
3.4.8 State creeping of paper
3.4.9 State the news print
3.4.10 State the grease proof ,glassine paper and, Tissue paper
3.4.11 Compare bond and currency papers
3.4.12 State Kraft paper and sack draft

MODULE IV

4.1.0 Understand Chemical Recovery operations In Paper Mills

4.1.1 Explain chemical recovery with flow diagram
4.1.2 State the black liquor and its properties
4.1.3 Describe the chemical composition of black liquor
4.1.4 Describe distillation and oxidation of black liquor
4.1.5 State the working of principles of evaporators
4.1.6 List the types of evaporators short tube evaporator, long tube evaporators, falling film evaporator, forced in circulation evaporator cascade evaporator disc evaporator
4.1.7 State the method of feeding in evaporators such as multiple effect and single effect
4.1.8 Calculate the steam economy in evaporator
4.1.9 State the different types of recovery boilers and furnaces
4.1.10 State the function and operations of recovery boilers such as Babcox and Wilcox

4.1.11 state the principles of causticizing process
4.1.12 State the function and chemical reactions of causticizing
4.1.13 State green liquor, white liquor, black liquor
4.1.14 Specify the reaction taking place during the process of black liquor and boiler and causticizing plant
4.1.15 State the functions and structure of lime shaker, classifiers, mud washers a lime debarring dryers
4.1.16 State lime reburning sludge disposal
4.1.17 List the chemical used in causticizing operation
4.1.18 Calculate the lime requirement in a lime reburning kiln
4.1.19 Describe the conventional sulphate recovery systems
4.1.20 State zeolite process and dimineralization process
4.1.21 Calculation of white liquor for cooking

4.2.0 Understand the Need and relevance of Effluent treatment and Air pollution control

4.2.1 state effluent treatment
4.2.2 State the need of effluent treatment
4.2.3 State chemical used in effluent plant- such as polyelectrolyte alum etc.
4.2.4 Describe the various effluents coming out of pulp and paper mills – such as air pollution, steam pollution, solid waste
4.2.5 Describe COD, BOD, and colour
4.2.6 State the method of air pollution control and their principles
4.2.7 State the equality control system in pulp mills

CONTENT DETAILS

MODULE I

Introduction – stock preparation- Defibration – Hydra Pulper – batch and continuous, Kneaders, Deflakers Storage chests and agitators – different types of stock and machine chests – agitator assemblies batch and continuous process of stock preparation Beating & Refining Beating & Refining processes – Theory & Beating and refining – factors influencing beating and refining – freeness – Hydration – Fibrillation – and their effects on paper properties different types of beaters & refiners, distinguish between beater & refiners Hollander beater- conical refiner, narrow wide angle shallow angle refiners single and double disc Consistency controllers and stock proportioning – different types of consistency controllers – working principles

MODULE II

Purpose of sizing – internal and surface sizing – sizing chemicals such as rosin, alum, and wax their preparation, theory of sizing – parameters influencing sizing , acidic alkaline and Neutral Sizing Surface sizing – Sizing chemicals starch as a sizing agent Size presses - straight horizontal, inclined and speed sizes Dyeing: Pigment dyes, Basic dyes, Acid dyes, Direct dyes, General Properties of different groups of dyes – different ways of adding the dye fastness of dye penetration of dyes into the fiber Factor affecting stock dyeing – Beating and refining - sizing – fillers, pH, order of addition – pulp characteristics Filling & Loadings: - Purpose of filling and loadings – different types of fillers, such as clay, talc, titanium dioxide calcium carbonate and magnesium carbonate Retention and effects of fillers on paper Properties – theory of retention – filtration. Co-flocculation – mechanical attachment Wet strength additives: Chemistry of additives influences of additives on cellulose –

different types of additives – natural and synthetic preparation and addition of wet strength additives Basis weight, caliper, density, bulk, folding endurance, tear strength, burst strength, tensile, stretch, paper testing – sample preparations surface properties – Roughness, smoothness pick strength, abrasion resistance oil absorbency, brightness, opacity, gloss sizing degree, colour, water absorbency stock proportioning – different methods and types Cleaning and screening – settling, centrifugal screens – HD and LD cleaning – Different methods of cleaning – pressure screens – centrifugal & cleaners – sorters Dearation – purpose – common type of dearators – decculators. Head box – head box and its parts – distribution section, middle section- slice straight, convergent and w type of head box Type of head boxes- open and closed – role of distributor roll – high turbulence head box – different type of manifolds – head box over Flow, head box re circulation, jet wire ratio, head calculations (simple problems only) Wire part – General arrangement and description of paper machine system and sheet formation type of paper machine wire part – cylinder mould- fourdrainer – twin wire formers – special former combinations Brief descriptions of different components of wire part foaming medium wire weave pattern sheet transfer factors affecting drainage and sheet formation

MODULE III

(a) Drying – principles of drying different methods of drying conduction convection and radiation pre drying factors- factors affecting drying – heat content temperature moisture different types of dryers and its arrangement yankee cylinder (MG). Multi cylinder (MF) combined – infrared radiator, flaket use of air in paper drying Steam and condensate removal – cascade – pressure regulated differential pressure regulated -– ejectors – Thermo compressors- Condensate removal method – bucket, scoop siphon, rotating and stationary. Rimming calculation of heat transfer, heat transfer rate Overall heat transfer coefficient – simple problems Dryer hoods open – closed high velocity exhaust fans, dryer felts and screens – its cleaning and drying permeability's Common troubles drying – psychrometry. Coating, calendering, reeling Coating – coating pigments – binders – coating formulations – coating equipment's, air knife coater, brush coater, cast coating, flexible blade cotter, gravure coating, knife coating reverse roll coater – transfer roll coater – curtain coating – calendering – theory of calendering – effect of calendering on paper quality open side closed side calender swimming rolls – Importance of moisture on paper during calendaring common calender troubles hard and soft spots sheet blackening cuts Finishing – converting – specialty – papers – finishing – winding different types of winder converting slitting sheet cutting rotary simplex duplex guillotines sorting counting packing –sheets – reels – roll wrapping machine weighing and labeling - scabs – reeling- pope reel – core mind reels – friction or surface drive reel

MODULE IV

Chemical recovery Black liquor – properties of black liquor – chemical composition of black liquor – deslication and oxidation Evaporators – working principles of evaporator, types of evaporators- long tube evaporators – short tube evaporator – falling film evaporator- forced circulation evaporators – multiple effects single effects, cascade evaporators. Disc evaporators Steam economy calculation in evaporators – recovery boiler and furnace Types - function, electrostatic, precipitator (ESP) reactions in recovery in boiler, make up chemical ID fan, FD fan, Boiler feed water properties Caustisizing Functions and chemical reaction batch and continuous process – green liquor classifiers and sledge removal Green liquor, white liquor, sledge, sulphidity, caustisizing lime slakes, classifiers, white liquor classifier mud washer operation techniques – lime reburning sludge disposal – make up chemicals – properties storage, transportation methods, handling and effect of lime impurities Brief discussion of conventional sulphate recovery systems, Zeolite process, demineralization process, simple calculation of white liquor for cooking Effluent treatment Pulp and paper mill effluent – physical and chemical nature of effluent and its treatment pollution control board and its norms chemicals used in effluent treatment, air pollution and its control methods

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